Electric Vehicle Management System Using Artificial Intelligence and Machine Learning

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Project Proposal Report

Predictive Charging & Breakdown Assistance System

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

The rapid adoption of Electric Vehicles (EVs) has created new challenges related to charging infrastructure availability and emergency assistance during vehicle breakdowns. Many EV users face difficulties locating available charging stations and managing low battery situations during travel. Existing navigation systems provide only static charging station information without predicting station availability or offering intelligent assistance during breakdown situations.

This research proposes a Predictive Charging & Breakdown Assistance System as part of an AI-based Electric Vehicle Management System. The system predicts charging station availability using historical usage data and recommends optimal charging stations with estimated waiting times. Additionally, it provides emergency assistance by identifying breakdown situations and generating battery-safe routes to the nearest service stations. Machine learning techniques and route optimization algorithms will be used to improve decision-making and ensure efficient EV operation.

The proposed system will integrate machine learning models, GPS location services, and charging station data to create an intelligent assistance platform for EV users. By predicting station congestion and providing emergency guidance, the system aims to reduce waiting time, improve travel efficiency, and enhance safety for EV drivers.

The expected outcomes include improved charging station utilization, reduced charging delays, and enhanced reliability of EV transportation systems. The system can contribute significantly to the sustainable transportation ecosystem by encouraging EV adoption and improving the user experience.

Keywords: Electric Vehicles, Charging Prediction, Machine Learning, Route Optimization, Breakdown Assistance

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
EV	Electric Vehicle
ML	Machine Learning
AI	Artificial Intelligence
GPS	Global Positioning System
API	Application Programming Interface

LIST OF APPENDICES

1. INTRODUCTION

1.1. Background and Context

The global transition toward electric vehicles (EVs) is accelerating as countries attempt to reduce carbon emissions and promote sustainable transportation. According to the International Energy Agency, global EV sales exceeded 14 million vehicles in 2023, showing a rapid increase in EV adoption worldwide. However, despite the benefits of EV technology, several operational challenges remain for EV users.

One major issue faced by EV drivers is the uncertainty of charging station availability. Drivers may arrive at charging stations only to find them occupied, leading to long waiting times and inefficient travel planning. Additionally, EV users often face difficulties when their vehicles experience breakdowns or extremely low battery levels during travel.

Existing navigation systems provide charging station locations but lack predictive capabilities and intelligent emergency assistance. These systems do not analyze historical usage patterns to predict station congestion or guide users safely during emergency situations.

To address these challenges, this research proposes an AI-based Predictive Charging & Breakdown Assistance System. The system will analyze charging station usage patterns, predict availability, recommend optimal charging locations, and provide emergency guidance when breakdown situations occur.

This research aligns with United Nations Sustainable Development Goal (SDG) 11: Sustainable Cities and Communities, which promotes sustainable transport solutions and efficient urban mobility systems.

1.2.Literature Review

Recent studies have explored the use of artificial intelligence and machine learning techniques to improve EV infrastructure management and charging efficiency. Several researchers have proposed predictive models to estimate charging demand and optimize charging station utilization.

Machine learning techniques such as regression models, neural networks, and time-series analysis have been used to forecast charging station usage patterns. These models help identify peak demand periods and improve station resource allocation.

Some navigation systems integrate geographic information systems (GIS) and real-time traffic data to optimize routes for EV drivers. However, most systems focus only on navigation and do not integrate predictive charging analytics.

Recent research has also explored anomaly detection techniques for EV systems, allowing the identification of abnormal vehicle behavior. However, integration of these technologies with charging station prediction and emergency assistance remains limited.

Although several EV support systems exist, many current solutions lack integrated predictive capabilities, real-time breakdown assistance, and personalized route recommendations based on battery conditions.

1.3.Research Gap

Although various EV support systems have been developed, several key challenges remain unaddressed. Many existing systems lack the ability to predict charging station availability using real-time and historical data. This limitation often results in long waiting times for EV drivers and inefficient use of charging infrastructure.

Furthermore, current EV navigation systems typically do not consider battery safety constraints when recommending routes to charging stations or service centers. As a result, drivers may receive navigation suggestions that are not feasible given their remaining battery capacity.

Another significant gap is the lack of emergency assistance mechanisms within EV navigation systems. Most systems do not provide intelligent guidance during breakdown situations or low battery emergencies.

Therefore, this research proposes a predictive assistance system that integrates charging station prediction, battery-aware route optimization, and emergency breakdown assistance. The system aims to enhance EV driver safety and improve charging infrastructure utilization through intelligent decision support.

2. OBJECTIVE

2.1.Main Objective

To develop an intelligent predictive system that recommends optimal EV charging stations and provides breakdown assistance using machine learning and route optimization techniques.

2.2.Specific Objectives

1. To analyze historical charging station usage data to predict station availability.
2. To recommend optimal charging stations with estimated waiting times.
3. To detect EV breakdown alerts or emergency situations from users.
4. To generate battery-safe navigation routes to nearby service stations.
5. To provide emergency driving guidance and safety recommendations for EV users.

3. METHODOLOGY

3.1. Research Approach

This research will follow a Design Science Research approach, where a technological solution is designed, implemented, and evaluated to address the identified research problem.

System Development Process

The development process will consist of the following stages:

1. Problem analysis
2. System design
3. Data collection
4. Machine learning model development
5. System implementation
6. Testing and evaluation

3.2. Technical Methods

The system will utilize the following techniques:

Machine Learning Models

- Time series prediction for charging station availability.
- Regression models for wait time estimation.

Route Optimization

- Shortest path algorithms using GPS and map services.

Emergency Detection

- User-triggered SOS alerts.

3.3. Tools and Platforms

Tool	Purpose
Python	Machine learning development
Node.js	Backend API development
MongoDB	Database management
Google Maps API	Route navigation
TensorFlow / Scikit-learn	Machine learning models
Firebase / AWS	Cloud infrastructure

3.4. Data Collection

Data sources may include:

- Charging station usage datasets
- EV battery level monitoring data
- Location and route data from GPS services

User data will be collected with consent and handled securely.

3.5. Validation Strategy

System performance will be evaluated using:

- Charging station prediction accuracy
- Route efficiency measurements
- Response time of emergency assistance system

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1. Overview of proposed solution and its components

The proposed system is an AI-driven Predictive Charging & Breakdown Assistance System designed to support electric vehicle (EV) users by providing intelligent charging recommendations and emergency assistance. The system integrates multiple components that work together to analyze user data, predict charging station availability, and ensure safe navigation during both normal and breakdown conditions.

The architecture consists of five main modules:

1. User Input Module – captures real-time user data such as location and battery level.
2. Charging Station Prediction Engine – analyzes historical and real-time data to forecast station availability and waiting times.
3. Route Optimization Module – calculates optimal and battery-safe routes using mapping services.
4. Breakdown Detection & Emergency Assistance Module – identifies emergency situations and provides immediate guidance.
5. Database & External Services Integration – manages data storage and integrates with APIs such as GPS and map services.

These components collaboratively deliver a seamless and intelligent EV assistance experience.

4.2.System diagram of components

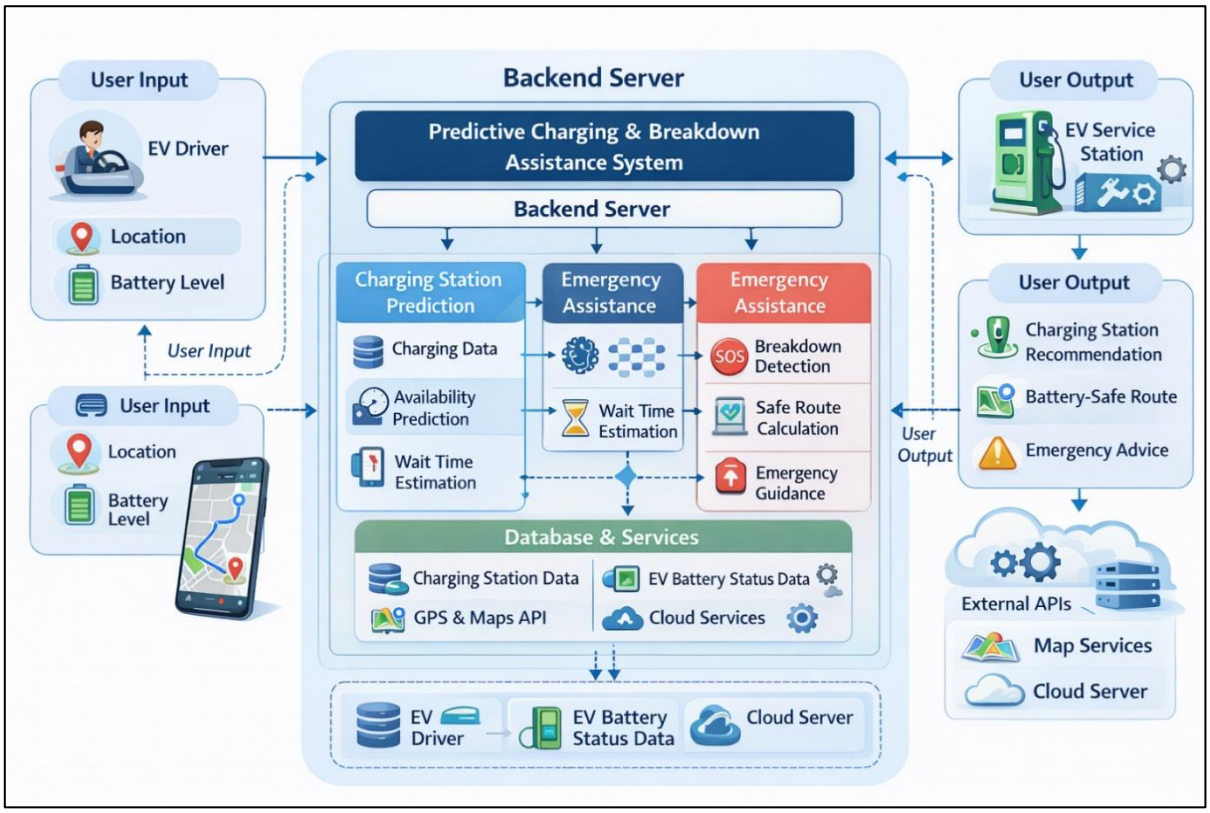


Figure 4.2: System Diagram of Component

The system diagram visually represents the interaction between the modules, illustrating how user inputs are processed through predictive and optimization engines to generate outputs such as charging recommendations and emergency routes.

4.3.Highlight the individual component and its parts

The Predictive Charging & Breakdown Assistance component is the core of the system and consists of two main sub-modules:

1. Charging Station Prediction Sub-module

- Availability prediction using historical charging data
- Waiting time estimation
- Station ranking based on proximity and demand

2. Breakdown Assistance Sub-module

- Breakdown detection (manual SOS or system-triggered alerts)
- Identification of nearest service stations
- Battery-safe route calculation
- Emergency guidance and safety recommendations

This component integrates machine learning and location-based services to provide intelligent and context-aware assistance to EV users.

4.4.Component interactions

The system modules interact in a structured and coordinated manner:

- The User Input Module sends real-time location and battery data to the backend system.
- The Charging Station Prediction Engine processes this data along with historical datasets to estimate station availability and waiting times.
- The Route Optimization Module uses prediction results along with GPS data to determine the best route.
- If a breakdown or low battery condition is detected, the Breakdown Assistance Module is triggered to provide emergency support.
- The Database and External Services Module supports all components by providing necessary data and API integrations.

These interactions ensure that the system responds dynamically to user needs and environmental conditions.

4.5.Data flow explanation

The system follows a structured data flow:

1. The user inputs location and battery level through a mobile application.
2. This data is transmitted to the backend server.
3. The prediction engine analyzes charging station data to forecast availability and waiting times.
4. The route optimization module calculates the best route to the selected charging station.
5. If a breakdown scenario occurs, the emergency module identifies the nearest service station and generates a battery-safe route.
6. Final outputs, including charging recommendations, navigation routes, and emergency alerts, are delivered to the user interface.

This data flow ensures efficient decision-making and real-time assistance for EV users.

4.6.Scalability and security considerations

Scalability

The system is designed to handle a growing number of users and data sources by utilizing:

- Cloud-based infrastructure for flexible resource allocation
- Distributed databases for efficient data management
- Scalable APIs to support real-time processing

This ensures that the system can expand as EV adoption increases.

Security

To protect user data and maintain system integrity, the following security measures are implemented:

- User authentication and authorization to prevent unauthorized access
- Data encryption for secure communication between client and server
- Secure cloud storage for sensitive data such as location and battery information
- Compliance with data privacy standards and ethical guidelines

These measures ensure that the system is both reliable and secure for real-world deployment.

5. USER REQUIREMENTS

5.1. Users/Stakeholder identification

- Electric vehicle drivers
- Charging station operators
- EV service providers

5.2. Functional requirements

1. The system shall collect user location and battery level data.
2. The system shall predict charging station availability.
3. The system shall recommend optimal charging stations.
4. The system shall provide wait time estimates.
5. The system shall detect breakdown alerts.
6. The system shall provide navigation routes to service stations.

5.3. Non-functional requirements

Security
User data must be encrypted and protected.

Performance
Charging recommendations must be generated within a few seconds.

Usability
The system must provide a simple and user-friendly interface.

Scalability
The system should support large number of EV users.

6. COMMERCIALIZATION PLAN

6.1. Target market and customer persona

The primary target market for the proposed system includes electric vehicle (EV) users, particularly those who rely on public charging infrastructure for daily commuting and long-distance travel. With the increasing adoption of EVs in countries like Sri Lanka and globally, there is a growing demand for intelligent systems that enhance charging efficiency and safety.

Secondary target customers include:

- EV fleet management companies (e.g., ride-hailing or delivery services)
- Charging station providers seeking to optimize station utilization
- Smart city planners and transportation authorities

Customer Persona

A typical user of this system would be:

- A working professional aged 25–50
- Owns or uses an electric vehicle regularly
- Travels frequently in urban or semi-urban areas
- Concerned about battery range, charging delays, and safety during travel

This user values convenience, reliability, and real-time assistance, making them an ideal customer for the proposed solution.

6.2. Value proposition

The proposed system delivers a unique value proposition by combining predictive analytics and emergency assistance into a single intelligent platform.

Key value offerings include:

- Reduced waiting time through accurate prediction of charging station availability
- Optimized travel planning with battery-safe route recommendations
- Enhanced safety through real-time breakdown detection and emergency guidance
- Improved user experience with personalized and data-driven recommendations

Unlike traditional navigation systems, this solution provides proactive decision-making support, enabling EV users to travel more efficiently and confidently.

6.3. Revenue model

The system will adopt a multi-stream revenue model to ensure sustainability and scalability:

1. Subscription Model
    - Basic features offered for free
    - Premium features (advanced predictions, priority recommendations, emergency services) available through monthly or yearly subscriptions
2. Partnership Model
    - Collaborations with charging station providers to promote their services
    - Revenue through advertisements or featured station listings
3. Enterprise Solutions
    - Licensing the system to EV fleet operators and logistics companies
    - Custom solutions for corporate clients
4. API Monetization
    - Providing access to prediction APIs for third-party developers and EV platforms

6.4. Cost estimation

The major costs involved in commercialization include:

- Development Costs – system design, machine learning model development, and application development
- Infrastructure Costs – cloud hosting, database management, and API usage
- Operational Costs – system maintenance, updates, and customer support
- Marketing Costs – promoting the application through digital marketing and partnerships

Initially, the system can be launched with a minimum viable product (MVP) to reduce costs, followed by gradual scaling.

6.5. Pricing strategy

The pricing strategy will follow a freemium model, allowing users to access basic services for free while charging for advanced features.

Proposed Pricing

- Free Tier
    - Basic charging station recommendations
    - Limited prediction accuracy
- Premium Tier (Rs. 500 – 1000/month)
    - Advanced prediction accuracy
    - Real-time wait time estimation
    - Emergency assistance features
    - Priority routing and notifications
- Enterprise Pricing
    - Custom pricing for fleet operators and business clients

This approach encourages user adoption while generating revenue from advanced services.

6.6. Competitive advantage

The proposed system offers several competitive advantages over existing solutions:

- Integrated Solution – combines charging prediction, route optimization, and emergency assistance
- AI-Driven Insights – uses machine learning for accurate predictions
- Battery-Aware Routing – ensures routes are feasible based on battery level
- Real-Time Emergency Support – provides immediate assistance during breakdown situations

These features differentiate the system from traditional navigation apps and standalone EV tools.

6.7. Intellectual property considerations

To protect the innovation and ensure long-term sustainability, the following intellectual property (IP) strategies will be considered:

- Software Copyright for the developed system and source code
- Patent Opportunities for unique predictive algorithms and routing mechanisms
- Trademark Registration for branding and product identity
- Data Protection Compliance to ensure user data privacy and legal compliance

Additionally, licensing agreements can be established when partnering with third-party organizations or service providers.

The commercialization plan demonstrates that the proposed system has strong potential for real-world adoption and revenue generation. By targeting EV users and industry stakeholders, offering a unique value proposition, and implementing a scalable business model, the system can successfully transition from a research project to a market-ready solution.

7. BUDGET AND JUSTIFICATION

Table 7: Budget and Justification

Category	**Item**	**Estimated Cost (Rs.)**
Hardware Costs	Laptop (Development & Testing)	250,000
	Mobile Device (Testing EV App)	60,000
	Internet & Networking	15,000
Software Licenses	Development Tools (IDE – mostly free, contingency)	10,000
	API Subscription (Google Maps / Map Services)	25,000
Cloud Services	Cloud Hosting (AWS / Firebase – 6 months)	30,000
	Database Hosting & Storage	15,000
Human Effort Cost	Development Effort (Student contribution – estimated)	150,000
	Testing & Maintenance Effort	40,000
Total Estimated Cost		**595,000 Rs.**

7.1.Justification for Each Expense

7.1.1. Hardware costs

- A laptop is required for system development, machine learning model training, and backend implementation.
- A mobile device is necessary to test the EV user application and ensure proper functionality of real-time features such as GPS and notifications.
- Internet and networking are essential for cloud access, API integration, and real-time data processing.

7.1.2. Software licenses

- Most development tools (e.g., VS Code, Python libraries) are free, but a budget is allocated for premium tools or plugins if needed.
- Map APIs (Google Maps / OpenStreetMap services) may require paid usage for route optimization and real-time navigation features.

7.1.3. Cloud services

- Cloud platforms such as AWS or Firebase are required to host the backend system and enable scalability.
- Database hosting is necessary to store charging station data, user information, and prediction results securely.

7.1.4. Human effort cost

- Although the project is developed by a student, an estimated cost is assigned to reflect the development effort, time, and technical contribution.
- Additional effort is required for testing, debugging, and system maintenance, ensuring reliability and performance.

The total estimated budget for the proposed system is approximately Rs. 595,000, which covers all essential resources required for development, deployment, and testing. The investment is justified by the system's potential to improve EV user experience, optimize charging infrastructure usage, and enhance safety through intelligent assistance.

8. WORK BREAKDOWN STRUCTURE (WBS)

8.1. Task breakdown per member

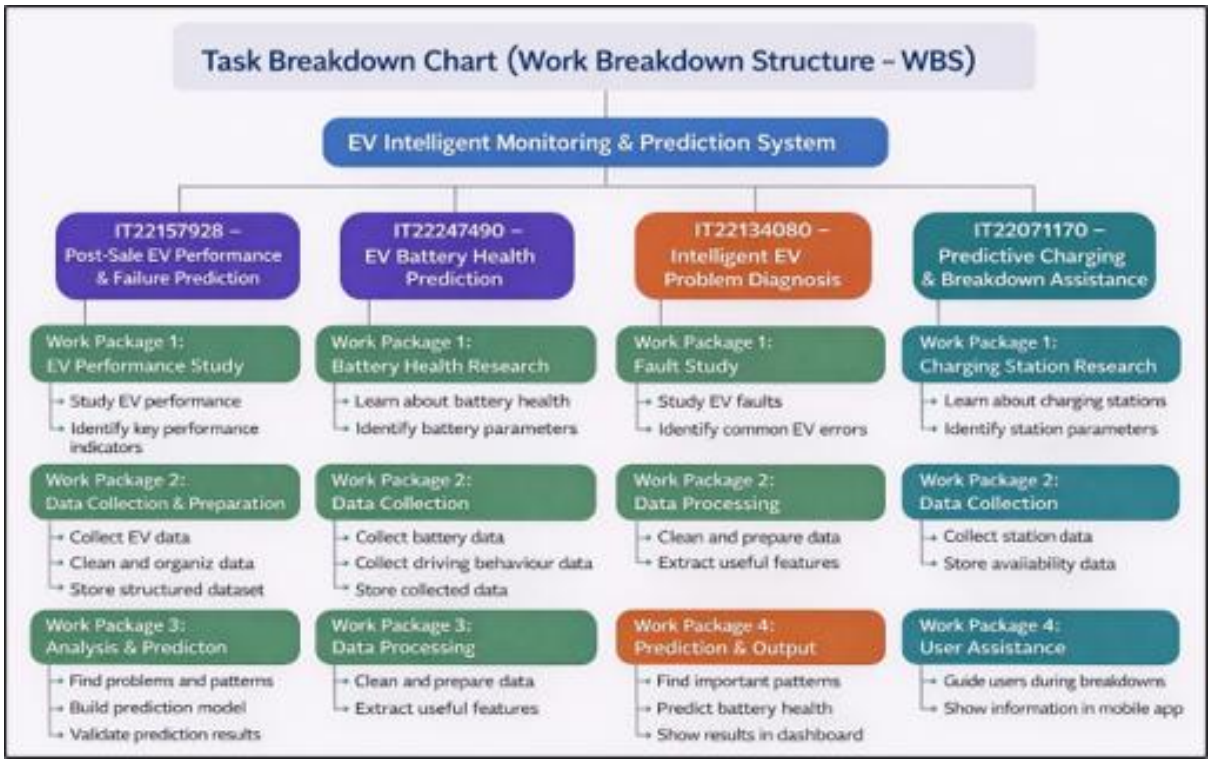


Figure 8.1: Task Breakdown Chart

The Work Breakdown Structure (WBS) outlines the division of tasks among project members, ensuring systematic development, proper time management, and balanced workload distribution.

8.2. Timeline alignment (Refer to Gantt Chart)

8.3. Balanced workload distribution

To ensure fairness and efficiency, the workload is distributed equally among all team members:

- Each member is responsible for one major system component, ensuring clear ownership and accountability.
- Tasks are divided into research, development, and testing, so every member contributes equally across different stages.
- Common tasks such as system integration, documentation, and presentation preparation are shared among all members.
- Regular team meetings and progress reviews ensure that no member is overloaded and all tasks are completed on time.

Additionally:

- Complex tasks such as machine learning model development and system integration are collaboratively handled when necessary.
- Workload is adjusted dynamically based on project progress and individual strengths.

9. GANTT CHART

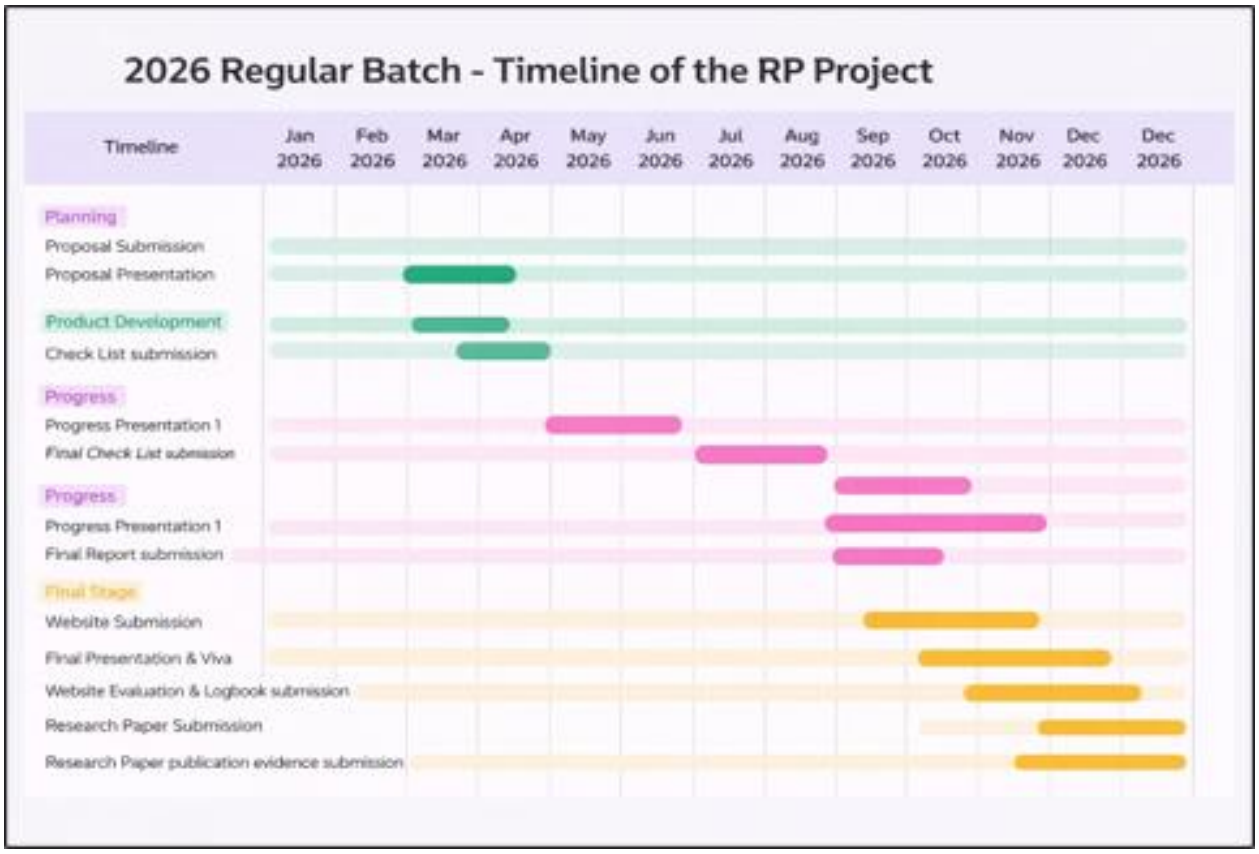


Figure 9: Gantt Chart

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11.APPENDICES

11.1. Survey instruments

A survey was designed to gather insights from potential EV users regarding their challenges with charging and breakdown situations.

Sample Survey Questions

Section A: General Information

1. Do you currently own or use an Electric Vehicle (EV)?
    - Yes / No
2. How frequently do you use your EV?
    - Daily / Weekly / Occasionally

Section B: Charging Experience

3. How often do you face difficulty finding available charging stations?

- Very Often / Often / Rarely / Never

4. How long do you typically wait at charging stations?
    - Less than 10 minutes / 10–30 minutes / More than 30 minutes
5. Would you prefer a system that predicts charging station availability?
    - Yes / No

Section C: Breakdown & Emergency Situations

6. Have you ever experienced a low-battery emergency while driving?

- Yes / No

7. How confident are you in finding nearby service stations during emergencies?
    - Very Confident / Somewhat / Not Confident
8. Would you use a system that provides emergency assistance and safe routing?
    - Yes / No

Section D: User Expectations

9. What features would you prefer? (Select all)

- Charging prediction
- Route optimization
- Emergency alerts
- Battery health monitoring

10. Any additional suggestions:

- (Open-ended)

11.2. Detailed diagrams

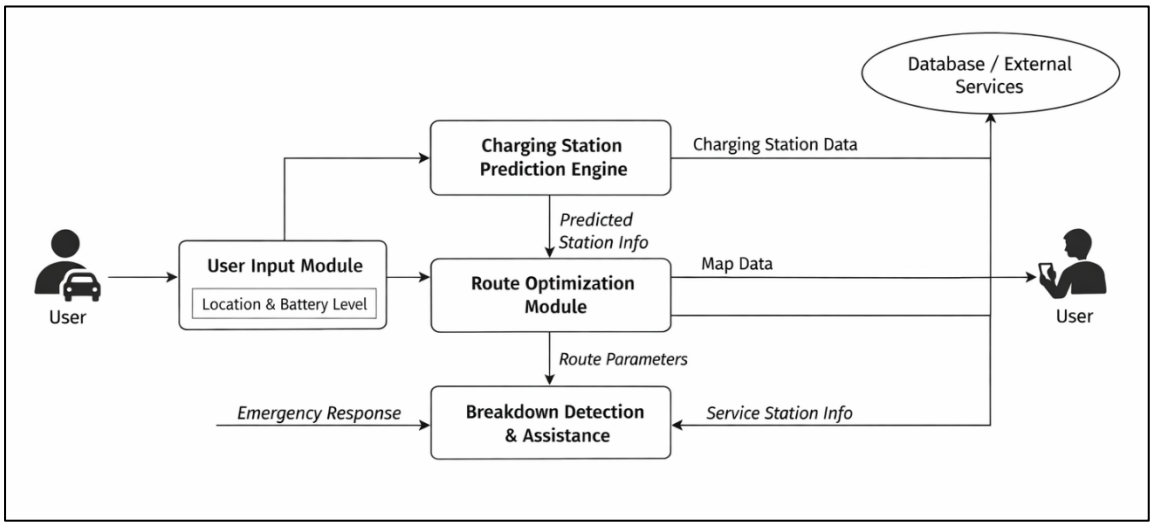


Figure 11.2: Data Flow Diagram (DFD) of the Predictive Charging & Breakdown Assistance System

These diagrams illustrate how different system components interact and how data flows between modules such as user input, prediction engine, and routing system.

11.3. Preliminary UI mockups

The following UI mockups represent the initial design of the system interface:
Key Screens:

1. User Dashboard
    - Displays battery level and current location
    - Shows nearby charging stations
2. Charging Recommendation Screen
    - Displays predicted availability
    - Shows waiting time and station details
3. Navigation Screen
    - Provides optimized route to charging station
4. Emergency Assistance Screen
    - Displays nearest service stations
    - Provides emergency instructions

These mockups ensure a user-friendly and intuitive interface for EV users.

11.4. Risk analysis

The project involves several potential risks that must be identified and mitigated.

Table 11.4: Risk Analysis

Risk	**Impact**	**Mitigation Strategy**
Inaccurate prediction results	Poor user experience	Improve model training and data quality
Limited data availability	Reduced system accuracy	Use public datasets and simulated data
API dependency failures	System downtime	Use backup APIs and caching
Security vulnerabilities	Data breaches	Implement encryption and authentication
System scalability issues	Performance degradation	Use cloud-based scalable infrastructure

11.5. Ethical documents

The system involves collecting and processing user data, which requires adherence to ethical standards.

Ethical Considerations:

- User Consent: Users must provide consent before sharing location and battery data.
- Data Privacy: Personal data will be securely stored and not shared with third parties without permission.
- Transparency: Users will be informed about how their data is used.
- Data Minimization: Only necessary data will be collected.

Ethical Approval

- The research will comply with institutional ethical guidelines.
- Any data collection involving users will follow proper approval procedures.